Diaphragm Pump Design & Simulation Workflow

Engineering Design Steps, Inputs & Simulation Pipeline

1. REQUIREMENTS & PERFORMANCE (Initial Sizing)

Key Inputs:

- Required Flow Rate Q
- Maximum Operating Pressure P_{max}
- Vacuum / Suction Conditions $P_{vac/suc}$

Analytical Calculations & LaTeX Equations:

- **Displacement Volume:** $V_d = \frac{Q}{N \cdot \eta_v}$
- **Stroke & Eccentric Radius:** $S = 2r \quad , \quad V_d = A_{eff} \cdot S$
- **Motor Power:** $P_{motor} = \frac{Q \cdot \Delta P}{\eta_{total}}$
- **Shaft Diameter:** $d \geq \left(\frac{16T}{\pi \tau_{allow}}\right)^{1/3}$
- **Valve Port Area:** $A_v = \frac{Q}{v_{max}}$

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2. MATERIAL SELECTION & STRUCTURAL FEA (ANSYS)

Inputs:

- Material Properties (EPDM, NBR, PTFE - Hyperelastic models)
- Boundary Conditions (Fixed outer rim, center deflection/force)

- **Hyperelastic Strain Energy Potential:** $W = W(I_1, I_2, I_3)$
- **Stress-Strain Relations:** $S_{ij} = \frac{\partial W}{\partial E_{ij}}$
- **Fatigue Life Prediction:** $N_f = C \cdot (\Delta\sigma)^{-\beta}$

Outputs / Evaluation:

- Stress & Strain Distribution
- Deflection Profile & Fatigue Life Analysis

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3. FLUID DYNAMICS (CFD)

Inputs:

- Fluid Properties (Density ρ, Viscosity μ)
- Dynamic / Moving Mesh – UDF in ANSYS FLUENT (Driven by diaphragm motion)

Formulas & Governing Equations:

- **Navier-Stokes Equations:** $\rho\left(\frac{\partial \vec{v}}{\partial t} + \vec{v} \cdot \nabla \vec{v}\right) = -\nabla P + \mu \nabla^2 \vec{v} + \vec{f}$
- **Continuity Equation:** $\nabla \cdot \vec{v} = 0$
- **Valve Pressure Drop:** $\Delta P_{valve} = \xi \cdot \frac{\rho v^2}{2}$

Outputs / Evaluation:

- Valve Pressure Drop & Flow Velocity Profiles
- Volumetric Efficiency Verification

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4. SYSTEM-LEVEL SIMULATION (MATLAB / Simulink - Simscape)

Block Components:

- **Electric Motor Equations:** $T_m = K_t \cdot I \quad , \quad V = IR + L\frac{dI}{dt} + K_e \omega$
- **Kinematic Linkage / Crank-Slider:** $x(t) = r\cos(\omega t) + \sqrt{l^2 - r^2 \sin^2(\omega t)}$
- **Check Valve Dynamics:** $m\ddot{x}_v + c\dot{x}_v + kx_v = A_v \Delta P(t)$
- **Instantaneous Mass Flow Rate:** $\dot{m}(t) = \rho \cdot A(t) \cdot \sqrt{\frac{2\Delta P(t)}{\rho}}$

Final Outputs:

- Characterization Curves
- Overall System Performance Optimization